

BTF Viewer — Newbie Analysis Workflows

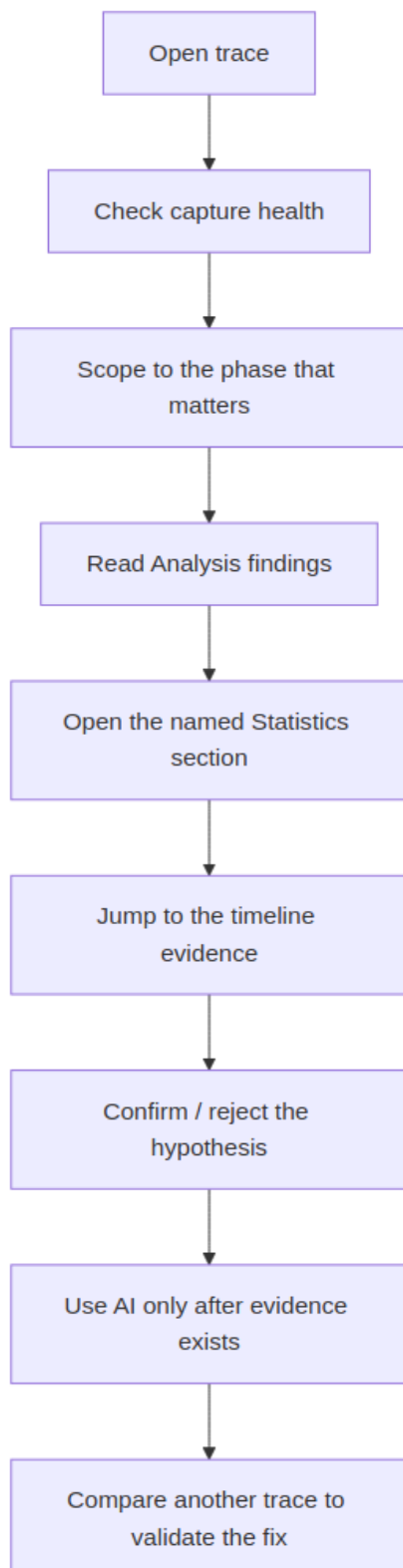
- Goal:** give a new BTFViewer user a repeatable way to find most RTOS scheduling issues step by step.
- Rule:** find deterministic evidence first, then ask AI to explain it.

This document is the procedure guide for diagnosing RTOS trace behavior with BTFViewer.

Document	Use it for
README.md	Product overview, UI tour, quick start, demo, export, settings
WORKFLOWS.md	Step-by-step diagnosis procedure — this document
STATISTICS.md	Metric definitions, formulas, interpretation, chart behavior
AI.md	AI Assistant architecture, model setup, tools, planner, validator, benchmark

0. Mental model

BTFViewer is evidence-first:



The AI Assistant is useful, but it is not the first source of truth. It explains **Analysis Findings, Statistics**, and **Trace Compare** data. It does not replace the deterministic BTF measurements.

1. Fast path: 10-minute triage

Use this when you open an unknown trace.

	Step	What to do	Why
	1	Open the trace and press Ctrl+0 / Fit	Confirm the whole capture is visible
	2	Turn on Load and switch between Task / Core view	Understand overall activity
	3	Open Statistics and toolbar Analysis	Overview (quality, incident clusters, phase window) plus severity-tagged findings
	4	Read Trace Health (TICK) first	Bad or tickless timebase changes how timing should be interpreted
	5	Open only the Statistics sections named by Analysis	Avoid chasing every table
	6	Click Max, p95 , a table row, chart point, or heatmap cell	Jump to the actual timeline evidence
	7	Place C1-Cn cursors around the phase and enable Limit to C1-Cn	Remove unrelated phases from the numbers

	Step	What to do	Why
	8	Re-read Analysis and Statistics inside that scope	Confirm the issue is still present
	9	Use Start Investigation , Investigate... , Verify with AI... , or Explain region (Ctrl+K opens Analysis / AI / workspace presets / Inspect task)	Ask AI to explain evidence, not invent it
	10	Export HTML or run Trace Compare after a fix	Keep a record and validate improvement

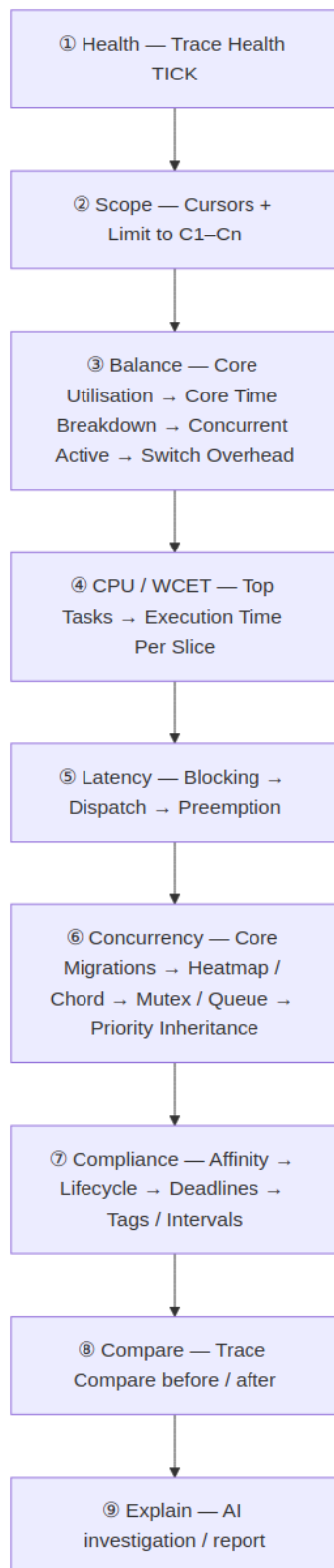
Stop rule

Stop drilling when one of these is true:

- The measured evidence explains the symptom.
 - The issue disappears after scoping to the relevant phase.
 - The required instrumentation is missing.
 - The next step is a firmware change and recapture.
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2. The top-down ladder

Most trace problems can be handled by this order:



Why this order works

Ladder rung	Question answered
Health	Can I trust timing metrics in this capture?
Scope	Am I looking at the right phase?
Balance	Is work distributed across cores?
CPU / WCET	Which task runs too long on CPU?
Latency	Which task waits too long, and why?
Concurrency	Is SMP scheduling causing migration, lock-bounce, or priority issues?
Compliance	Did affinity, suspend/resume, deadlines, and custom signals behave as expected?
Compare	Did a change actually improve the measured behavior?
Explain	Can AI summarize and challenge the evidence?

3. Step-by-step workflow

Step 1 — Check trace health**Open: Statistics → Trace Health (TICK)**

Check	If you see this	Do this
TICK mode, low CV	Timing source looks stable	Continue

Check	If you see this	Do this
TICKLESS, high CV	Idle tick suppression may be expected	Scope to a busy window and re-check
Large gaps during busy work	Possible lost ticks, long critical section, or trace gap	Inspect the timeline and compare with Execution / Blocking
No useful TICK evidence	Timing interpretation is weaker	Prefer relative comparisons and explicit interval/tag instrumentation

Newbie rule: do not treat a tickless warning as a bug until you scope to a busy region.

Step 2 — Scope the phase

Mixed traces often contain setup, stress tests, sleeps, and teardown. Whole-trace numbers may hide the real issue.

Action	Result
Click timeline / press C	Place cursors
Ctrl+R	Zoom to cursor range
Enable Limit to C1-Cn	Recompute Statistics and Analysis for that phase
Clear cursors	Return to full-trace statistics

Important: the Migration **Heatmap / Chord** inspector follows the visible viewport, not the cursor-scope checkbox. Zoom the timeline before opening the inspector.

Step 3 — Check core balance**Open:** Statistics → **Core Utilisation**

Signal	Meaning	Next
Load Balance Score $\geq$ 85% and $\sigma \leq 30\%$	Generally balanced (Analysis still shows Score/ σ /G; “reasonably balanced”)	Continue to CPU / latency
Score < 70%	Imbalance likely	Check Affinity and Task × Core
$\sigma > 30\%$	Utilization spread is high	Check Core Time Breakdown
One core hot, others idle	Work may be pinned	Check Core Affinity
All cores high, one underused	Serialization or lock contention	Check Mutex / Preemption

Then open:

Section	What it reveals
Core Time Breakdown	Active / Idle / Tick / Gap time per core
Concurrent Core Active	How often N cores are busy together
Kernel Switch Overhead	Gap between one slice ending and the next starting
Task × Core	Which task used which core

Step 4 — Find CPU / WCET problems**Open:** Statistics → **Top Tasks by CPU**, then **Execution Time Per Slice**

Check	Red flag	Next
Top CPU tasks	One task dominates CPU	Inspect task behavior and deadline
Equal-priority workers dominate	Scheduler stress or fan-out issue	Consider affinity or reducing concurrency
Execution Max much larger than p95	Rare spike	Click Max and inspect the timeline
Execution Max exceeds budget	WCET concern	Set Deadlines / CPU budget thresholds
Long slice overlaps lock or preemptor activity	Interference	Open Mutex / Preemption

Do not rely only on Average. Use Max, p95/p99, and the distribution chart.

Step 5 — Diagnose latency

Latency questions usually split into three different metrics.

Metric	Meaning	Start here when...
Blocking Time	Off-CPU gap until the task runs again	A task waits too long
Dispatch / Scheduling Latency	Ready/create/resume → first run	A ready task does not run soon enough
Inter-Arrival / Period / Jitter	Start-to-start cadence	A periodic task is late or irregular
Response Time	Heuristic adjacent-slice ready→completion	You need a rough tail signal, not kernel release/completion

Workflow

1. Open **Blocking Time**, sort by **Max** or **p95**.
2. Ignore known orchestrator sleeps and setup/teardown phases.
3. Click the long gap and inspect the timeline.
4. Open **Preemption Chain** to see who ran while the task waited.
5. Open **Mutex / Semaphore / Queue** if the gap aligns with a hold.
6. Open **Dispatch / Scheduling Latency** only when the trace has create/resume STI evidence.

Important: BTFViewer Blocking Time is not true end-to-end response time. For true application response time, instrument release/completion with intervals or tags.

Step 6 — Diagnose SMP concurrency issues

This catches many multi-core problems: thrash, lock-bounce, queue sharing, and priority inheritance.

6.1 Core migrations **Open:** Statistics → **Core Migrations**

Signal	Meaning
High Rate	Task moves often relative to active time
Short Dwell	Task does not stay on one core long
High Ping	A→B→A bouncing
High Gap after	Migration may be followed by extra waiting
High STI±	Migration near software trace events

Procedure

1. Sort by **Rate**, **Dwell**, or **Ping**.
2. Click the row and inspect **Dwell / Rate / Gap** charts.
3. Switch to **Task View**, lock-highlight the task, enable **Load**.
4. Check whether the task is bouncing between cores or simply spreading.

6.2 Core-pair and heatmap **Open:** Statistics → **Core-Pair Migration Summary**, then toolbar **Heatmap / Chord**

Signal	Meaning
High pair count	Hot migration corridor
High Bounce %	Task migrated while holding a sync object
Hot heatmap cell	Burst of migrations in one time window
Hot chord ribbon	Heavy directed core-to-core movement

Use **Lock Bounces Only** when chasing mutex/queue-related hops.

6.3 Mutex / semaphore / queue **Open:** Statistics → **Mutex / Semaphore** or **Queue**

Signal	Meaning
Issues > 0	Pairing problem, teardown, cross-task give, unmatched take/give
Bounces high	Object crossed core boundaries while held
Long hold	Contention risk
Warning / Error	Inspect the timeline before acting

Typical fixes:

- Co-locate producer/consumer tasks.
- Pin tasks that share hot locks.
- Shorten critical sections.
- Reduce equal-priority runnable fan-out.

6.4 Priority inheritance **Open:** Statistics → **Priority Inheritance**

Pattern	Reading
Mutex inherit	Kernel boost is working
L/M/H pattern	Classic priority inversion geometry; review design
Boost only	Manual priority change or incomplete evidence

Click a boost episode or red timeline stripe to verify.

Step 7 — Compliance checks

Use this rung to verify that configured behavior actually happened.

Section	Check
Core Affinity	Observed cores should fit the active affinity mask
Task Lifecycle	Suspend/resume counts should match; task should not run while suspended
Deadlines / CPU budget	Violations should match your real budgets
Tag Analysis	Custom values should stay inside application limits
Interval Analysis	Instrumented regions should meet duration budgets

Deadline values are configured in **nanoseconds** under Settings → Display → Analysis thresholds.

Step 8 — Compare before / after

A fix is not validated until a second trace proves it.

Open: two traces → toolbar **Compare**

Compare page	What to check
Summary	Span, context switches, migrations, load balance, tick health
Core Util	Per-core utilization delta
Execution	WCET / p95 / Max changes
Blocking	Wait-time changes
Preemption	Whether interference improved
Sync / Mutex	Holds, issues, bounces
Response	Heuristic P99 signal
Trends	All open tabs (3+): load balance, migrations, tick health
Deadline misses	Pass/fail thresholds

Comparison rule: use equivalent workload phases. Place cursors in both traces and enable compare cursor scope when needed. **Save as baseline** on Trace A, then **Score vs baseline** after a recapture.

Step 9 — Use AI after deterministic evidence

Use AI when:

- Analysis Findings exist.
- Statistics are scoped correctly.
- You have clicked at least one timeline event.
- You want a narrative, hypothesis ranking, or experiment plan.

AI action	Best use
Triage findings / Analysis Findings	Summarize top issues
Explain region	Explain a cursor-scoped phase
Investigate...	Rank hypotheses and gather evidence
Verify with AI...	Confirm or reject a selected finding
Root cause...	Walk the top finding through related metrics
Start Investigation / Auto investigate...	Empty AI log (also after restart with no user/assistant turn) or full tool-driven drill-down
What-if / Optimize	Estimate possible experiments
Diagnostic report	Write a structured summary

Trust rule: click every jump:TIME and verify it on the timeline. Treat What-if / Optimize as estimates, not measured results. After **Clear**, replies, usage cost, and current investigation issues are gone (no leftover **Current Issue** card). Restart does not restore **Current Issue** unless the log still has a user or assistant turn — **Start Investigation** stays available.

4. Symptom → metric map

Symptom	Start here	Then check
Unknown problem	Toolbar Analysis	Statistics sections named by findings
Tick irregularity	Trace Health (TICK)	Scope busy window; Tick Distribution
Uneven SMP load	Core Utilisation	Task × Core, Affinity, Concurrent Active

Symptom	Start here	Then check
Rarely N cores active	Concurrent Core Active	Worker count, affinity, locks
High switch cost	Kernel Switch Overhead	Core Time Breakdown Gap %
Task burns CPU	Top Tasks by CPU	Execution Time Max / p95
Task slice too long	Execution Time Per Slice	Preemption, Mutex, Deadlines
Task waits too long	Blocking Time	Preemption Chain, Mutex
Ready task delayed	Dispatch / Scheduling Latency	create/resume STI evidence
Periodic task irregular	Period / Jitter	Inter-Arrival, Tick Health
Core thrashing	Core Migrations	Rate, Dwell, Ping, Heatmap
Lock-bounce	Core-Pair Bounce %	Heatmap Lock Bounces Only, Mutex Bounces
Priority inversion	Priority Inheritance	Mutex, Blocking, timeline stripes
Affinity wrong	Core Affinity	Task × Core, migrations
Suspend/resume issue	Task Lifecycle	Timeline STI
Custom latency	Interval / Tag Analysis	Instrumentation design
Regression after change	Trace Compare	Same scoped phase on both traces
Need explanation	AI Assistant	Verify timeline evidence first

5. Newbie checklist

Before concluding:

- Trace opened and full span checked
 - Statistics + Analysis opened
 - Trace Health checked first
 - Relevant phase scoped with C1–Cn if needed
 - Analysis finding mapped to a Statistics section
 - Max / p95 / row / chart point clicked on timeline
 - Core balance checked
 - CPU / WCET checked
 - Blocking / dispatch / preemption checked for latency issues
 - Migrations / Heatmap / Mutex / Priority Inheritance checked for SMP issues
 - Affinity / Lifecycle / Deadlines / Tags checked
 - AI used only after deterministic evidence exists
 - Any jump:TIME verified on timeline
 - Before/after fix validated with Trace Compare or exported report
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6. Worked example summary: **example-8cores**

The sample trace is intentionally noisy because it concatenates stress tests. Scope to one phase before treating a warning as a product defect.

Ladder rung	Example observation	Reading
Health	TICKLESS, CV ~36.7%, missed ~10	Expected in idle/tickless phases; scope busy window
Balance	Load Balance Score ~95%, σ ~6.3%	Generally balanced
CPU / WCET	CS workers around 15% CPU; Max slices ~3–4 ms	Equal-priority stress; inspect long slices
Latency	High task max block ~53 ms in inversion demo	Scope inversion phase; verify preemption/mutex
Migrations	~19k migrations; hot CS tasks ~1.6k/s	Scheduler thrash
Lock-bounce	Hot queue with hundreds of bounces	Co-locate producers/consumers

Ladder rung	Example observation	Reading
Priority	Low task boosted by priority inheritance	Kernel inheritance working
Compliance	Affinity and lifecycle checks pass	Keep as CI checks
Verdict	Thrash + queue bounces are P0	Pin/co-locate/reduce fan-out, then compare

7. Export and CI loop

After you find an issue:

```
python builds/btf_viewer.py report ../tracedata/example-8cores.btf.gz \
  --output report.html --format html
```

```
python builds/btf_viewer.py compare before.btf.gz after.btf.gz \
  --output compare.html --format html \
  --name-a "Before" --name-b "After"
```

Recommended loop:



8. What to read next

Need	Read
Basic UI, demo, settings, export	README.md
API keys	README.md#ai-api-keys
Metric definitions and formulas	STATISTICS.md

Need	Read
AI setup, model choice, tools, validator	AI.md
Repeatable practical diagnosis	This document
Guided walkthrough	README.md#demo